Project Title

Hospital Bed / OR Scheduling

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Hospital Bed and Operating Room Scheduling Using Advanced Algorithms

Efficient scheduling of hospital beds and operating rooms is a critical challenge in modern healthcare, where limited resources must be allocated while maintaining high-quality patient care. This project proposes an intelligent scheduling system that assigns patients and surgical procedures to available beds, operating rooms, and time slots based on multiple constraints, including treatment duration, patient priority, staff availability, resource capacity, and scheduling deadlines. Since hospital scheduling is a complex optimization problem that is computationally difficult to solve optimally for large datasets, the proposed system combines network flow techniques for resource allocation with approximation strategies to generate high-quality schedules within practical execution time.

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The scheduling engine aims to maximize resource utilization, reduce patient waiting time, minimize operating room idle time, and decrease the overall completion time (make span) while satisfying all operational constraints. The system also generates detailed reports containing bed occupancy, operating room utilization, scheduling efficiency, and constraint violations to support hospital administrators in decision-making. This project demonstrates the practical application of advanced data structures and algorithms, particularly network flow, assignment algorithms, and approximation techniques, to solve real-world healthcare resource optimization problems.

Signature of the Guide